

习题7

17. 证明拉普拉斯方程的第二边值问题

$$\begin{cases} u_{xx} + u_{yy} = 0 & 0 < x < a, \\ u_x(0, y) = f_1(y), \quad u_x(a, y) = f_2(y), \quad 0 \leq y \leq b, \\ u_y(x, 0) = g_1(x), \quad u_y(x, b) = g_2(x), \quad 0 \leq x \leq a \end{cases}$$

有解的必要条件是

$$\int_0^a [g_1(x) - g_2(x)] dx + \int_0^b [f_1(y) - f_2(y)] dy = 0$$

证 设该第二边值问题有解 $u(x, y)$, 并记矩形区域

$$\Omega = \{(x, y) \mid 0 < x < a, 0 < y < b\}$$

由方程 $u_{xx} + u_{yy} = 0$, 对整个矩形区域积分, 得

$$0 = \iint_{\Omega} (u_{xx} + u_{yy}) dx dy$$

分别对两项积分

$$\iint_{\Omega} u_{xx} dx dy = \int_0^b \left[\int_0^a u_{xx} dx \right] dy = \int_0^b [u_x(a, y) - u_x(0, y)] dy$$

$$\iint_{\Omega} u_{yy} dx dy = \int_0^a \left[\int_0^b u_{yy} dy \right] dx = \int_0^a [u_y(x, b) - u_y(x, 0)] dx$$

代入边界条件

$$u_x(0, y) = f_1(y), \quad u_x(a, y) = f_2(y),$$

$$u_y(x, 0) = g_1(x), \quad u_y(x, b) = g_2(x),$$

$$\text{于是, } 0 = \int_0^a [g_1(x) - g_2(x)] dx + \int_0^b [f_1(y) - f_2(y)] dy = 0$$